

#3. Basis for a Topology.

$$\#1. A = \bigcup_{x \in A} U_x.$$

#3. The Co-Countable is Topology, but τ_{co} is Not Topology.
[Counter example.

Let $X = \mathbb{R}$. Let $U_x = \mathbb{R} \setminus \{x\}$. Then, $U_x \in \tau_{co}$, but for fixed $y \in \mathbb{R}$,

$$\bigcup_{x \neq y} U_x = \mathbb{R} \setminus \{y\} \notin \tau_{co}.$$

That is, τ_{co} is Not closed under the union.

#4. a). Let $\{\tau_\alpha\}$ be a family of Topologies on X .

Then, $\bigcap \tau_\alpha$ is also Topology on X . But $\bigcup \tau_\alpha$ is not.

Proof. Clearly $\emptyset, X \in \bigcap \tau_\alpha$. And, let $U_i \in \bigcap \tau_\alpha$ ($i \in \mathbb{Z}$) be given

Then, $U_i \in \tau_\alpha$ for all $i \in \mathbb{Z}$ and $\alpha \in \Lambda$, then $\bigcup_{i \in \mathbb{Z}} U_i \in \tau_\alpha$ for all $\alpha \in \Lambda$.

So it is contained in $\bigcap \tau_\alpha$. Similarly, closed under the 'intersection'.

Counter example: $X = \{a, b, c\}$, $\tau_1 = \{\emptyset, X, \{a\}, \{a, b\}\}$, $\tau_2 = \{\emptyset, X, \{a\}, \{b, c\}\}$,

Then, $\{a, b\} \cap \{b, c\} = \{b\} \notin \tau_1 \cup \tau_2$

b). Let $S = \{U \mid U \in \tau_\alpha \text{ for some } \alpha\}$. Then, it is subbasis for X ,

and generates the smallest topology on X containing all τ_α .

The unique largest topology is $\bigcap \tau_\alpha$.

#5. If A is a basis of (X, τ) , then $\tau_A = \bigcap_{A \subseteq \tau'} \tau'$
Proof. Let $U \in \tau_A$ be given.
 τ' is topology.

Then, for some $\mathcal{O} \subseteq A$, $U = \bigcup \mathcal{O} \in \bigcap \tau'$.

The converse is trivial.

#6. Let $(-1,1) \setminus K \in \mathcal{U}_K$ be given. Then, $(-1,1) \setminus K \notin \mathcal{U}_I$, because for $0 \in (-1,1) \setminus K$, there is no basis element of $\mathcal{U}_I$ s.t. $0 \in B \subseteq (-1,1) \setminus K$. Conversely, $[0,1) \in \mathcal{U}_I$ is not contained in $\mathcal{U}_K$, because for $0 \in [0,1)$, there is no basis element of $\mathcal{U}_K$ s.t. $0 \in B \subseteq [0,1)$.

#7,

#8. b) The given $\mathcal{C}$ cannot generate $[\sqrt{2}, 3)$.

If $[\sqrt{2}, 3) \in \mathcal{U}_{\mathcal{C}}$, then for some $a_i \in \mathbb{Q}$ ($i \in \mathbb{Z}$),

$$[\sqrt{2}, 3) = \bigcup_{i \in \mathbb{Z}} [a_i, b_i)$$

Then, $\sqrt{2} \in [a_j, b_j)$ for some $j \in \mathbb{Z}$, and since $a_j \in \mathbb{Q}$, $a_j < \sqrt{2}$.

Contradiction.

Subspace.

#5. Let $(X, \mathcal{C}_1)$ and $(X, \mathcal{C}_2)$ be Topological spaces with $\mathcal{C}_1 \subseteq \mathcal{C}_2$ and let $(Y, \mathcal{C}_3)$ and $(Y, \mathcal{C}_4)$ be Topological spaces with $\mathcal{C}_3 \subseteq \mathcal{C}_4$. Then, $(X, \mathcal{C}_1) \times (Y, \mathcal{C}_3)$ is coarser than $(X, \mathcal{C}_2) \times (Y, \mathcal{C}_4)$.

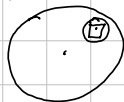
a) is trivial. To show the converse, let $U \in \mathcal{C}_2$ be given.

Then, $U \times Y \subseteq X \times Y$ is open, and $X' \times Y'$ is finer than $X \times Y$, $U \times Y \subseteq X' \times Y'$ is open.

Now, $\pi[U \times Y] = U \subseteq X'$ is open. $\square$

#6. Let $B_r(x) \subseteq \mathbb{R}^2$ be given. Then, for any $y \in B_r(x)$, put

$$\epsilon = r - d(x, y).$$



Finally, take $a, b, c, d \in \mathbb{Q}$ s.t.

$$y_1 < a < y_1 + \epsilon/\sqrt{2}, \quad y_1 - \epsilon/\sqrt{2} < b < y_1$$

$$y_2 < c < y_1 + \epsilon/\sqrt{2}, \quad y_2 - \epsilon/\sqrt{2} < d < y_2.$$

#7. Let $X = \mathbb{R} \setminus \{0\}$, $Y = (0, \infty)$.

Then Y is convex, but neither ray nor interval.
($0 \notin Y$)

#8. If L is a graph of $x = a$ for some $a \in \mathbb{R}$,

then $L \cong \mathbb{R}$ under the subspace of $\mathbb{R}_d \times \mathbb{R}$,

and $L \cong \mathbb{R}_d$ " $\mathbb{R}_d \times \mathbb{R}_d$.

And, in another case, $L \cong \mathbb{R}_d$.

#9.

Let $\mathcal{U} \subseteq \mathbb{R}^2$ be an basis of $\mathbb{R}^2$ with dic. order.

Let $(x, y) \in \mathcal{U}$ be given. Since $\mathbb{R}_d$ is discrete, $\{x\}$ is open of $\mathbb{R}_d$.

Meanwhile, the $\mathcal{U} = \{(a, b), (c, d) \mid (a, b) < (c, d)\}$, for this y ,

there is $\varepsilon > 0$ s.t. $(x, y - \varepsilon) < (x, y) < (x, y + \varepsilon)$ and $(x, y \pm \varepsilon) \in \mathcal{U}$.

Now, $(x, y) \in \{x\} \times (y - \varepsilon, y + \varepsilon) \subseteq \mathcal{U}$, thus $\mathbb{R}_d \times \mathbb{R}$ is finer than $\mathbb{R}^2$, dic.

Conversely, let $\{x\} \times (c, d) \subseteq \mathbb{R}_d \times \mathbb{R}$ be a basis element.

Then, it is clearly open of $\mathbb{R}^2$ dic. order.

#10. $\mathbb{I}^2 \neq \mathbb{I}_d^2$, because $(\frac{1}{2}) \times (\frac{1}{3}, \frac{2}{3})$ is open of $\mathbb{R}_d$, but not open in $\mathbb{R}^2$.

But, any basis of $\mathbb{I}^2$ is open of $\mathbb{I}_d^2$, because

$$(x_1, x_2) \in \mathbb{B}_r(x) \cap \mathbb{I}^2,$$

there is $\varepsilon > 0$ s.t. $(y_1, y_2) \in ((x_1, y_2 - \varepsilon), (x_1, y_2 + \varepsilon)) \cap \mathbb{I}^2 \subseteq \mathbb{B}_r(x) \cap \mathbb{I}^2$.

Section 17. Closed sets and Limit Points

#2. If A is closed in Y and Y is closed in X , then A is closed in X .

Proof. Y is closed in $X \Leftrightarrow X \setminus Y$ is open of X

A is closed in $Y \Leftrightarrow Y \setminus A$ is open of Y

$\Leftrightarrow Y \setminus A = Y \cap U$ for some open $U \subseteq X$

$X \setminus A = (X \setminus Y) \cup (Y \setminus A) = (X \setminus Y) \cup (Y \cap U) = (X \setminus Y) \cup U$ is open of X .

#3. If $A \subseteq X$ closed and $B \subseteq Y$ closed, then $A \times B \subseteq X \times Y$ is closed.

Proof. $(X \times Y) \setminus (A \times B) = (X \setminus A) \times Y \cup X \times (Y \setminus B)$

#4. Let Y be a Hausdorff space, X be any space, and $A \subseteq X$ be dense in X .

If $f, g: X \rightarrow Y$ are continuous and $f|_A = g|_A$, then $f = g$.

Proof. Let $x \in X$ be given. Suppose that $f(x) \neq g(x)$. Then, there are open $U, V \subseteq Y$

$f(x) \in U$, $g(x) \in V$, $U \cap V = \emptyset$.

Meanwhile, since f and g both are continuous, $f^{-1}[U]$ and $g^{-1}[V]$ are opens containing x .

Therefore, $f^{-1}[U] \cap g^{-1}[V] \cap A$ is non-empty open set, thus

$$f^{-1}[U] \cap g^{-1}[V] \cap A \neq \emptyset$$

since A is dense in X . Let $a \in f^{-1}[U] \cap g^{-1}[V] \cap A$. Then,

$f(a) \in U$, $g(a) \in V$, $a \in A$, thus $f(a) = g(a)$, $U \cap V \neq \emptyset$.

It is Contradiction.

#5. $(a,b) \subseteq [a,b]$ is trivial so enough to show that $(a,b)' \subseteq [a,b]$.

Let $\alpha \in X$ be a limit point of (a,b) . That is, for any open U containing α ,
 $(U \setminus \{\alpha\}) \cap (a,b) \neq \emptyset$. (*)

If $\alpha \notin [a,b]$, that is, $\alpha < a$ or $\alpha > b$. Either $\alpha < a$ or $\alpha > b$, see then:

$\alpha < a \Rightarrow [m, a)$ (if $m \in X$ is the smallest element) contains α , but does not satisfy (*).

" $\Rightarrow (r, a)$ (if such m does not exist) $\neq$

And similarly in case of $\alpha > b$.

The equality cannot hold when $X = \mathbb{N}$, $(1, 3)$.

#6. a) is clear, by definition.

b) is shown by:

$x \in \overline{A \cup B} \Leftrightarrow$ for any open U containing x , $U \cap (A \cup B) \neq \emptyset$

and since $U \cap (A \cup B) = (U \cap A) \cup (U \cap B)$, either $U \cap A$ or $U \cap B$ is non-empty.
So, $x \in \overline{A}$ or $x \in \overline{B}$. The converse is trivial by a).

c). For each B , $A_p \subseteq \bigcup A_\alpha$, so $\overline{A_p} \subseteq \overline{\bigcup A_\alpha}$.

But the converse is false, by the counterexample: $X = \mathbb{R}$, $A_n = \{\frac{1}{n}\}$. Then,

$$\bigcup_{n \in \mathbb{N}} A_n = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\} \text{ but } \bigcup_{n \in \mathbb{N}} \overline{A_n} = \{\frac{1}{n} \mid n \in \mathbb{N}\}.$$

#8. a), $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$. Let $X = \mathbb{R}$, $A = (0, 1)$, $B = (1, 2)$. Then

$$\overline{A \cap B} = \overline{\emptyset} = \emptyset \text{ but } \overline{A} \cap \overline{B} = \{1\}.$$

And, since $\overline{A \cap B} \subseteq \overline{A}$, $\overline{B}$, so $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.

b) has the same result.

c). Let $X = \mathbb{R}$, $A = [0, 1)$, $B = (0, 1)$. Then

$$\overline{A - B} = \{0\} \text{ but } \overline{A} - \overline{B} = \emptyset. \text{ So } \overline{A - B} \subset \overline{A} - \overline{B} \text{ does not hold.}$$

And, let $x \in \overline{A - B}$ be given. That is, $x \in \overline{A}$ but $x \notin \overline{B}$.

Let U be an open containing x s.t. $U \cap B = \emptyset$. Since $x \in \overline{A}$, $U \cap A \neq \emptyset$.

So, $U \cap (A - B) = (U \cap A) \cap (U \cap B^c) = (U \cap A) \cap (U - B) = U \cap A \neq \emptyset$

Thus $\overline{A - B} \subseteq \overline{A - B}$.

#9.

$(x, y) \in \overline{A} \times \overline{B} \Leftrightarrow$ for any open $U \subseteq A$, $V \subseteq B$ containing x, y , respectively, $U \cap A \neq \emptyset$ and $V \cap B \neq \emptyset$.
 $\Leftrightarrow$ for any basis element $U \times V \subseteq X \times Y$ containing (x, y) , $(U \times V) \cap (A \times B) \neq \emptyset$
 $\Leftrightarrow (x, y) \in \overline{A \times B}$.

#11. Let X and Y be Hausdorff spaces.

Given distinct $(x_1, y_1), (x_2, y_2) \in X \times Y$, either $x_1 \neq x_2$ or $y_1 \neq y_2$ must be hold
 WLOG, suppose $y_1 \neq y_2$. So, there are opens $U_1, U_2 \subseteq Y$ s.t.

$$U_1 \cap U_2 = \emptyset, \quad y_1 \in U_1, \quad y_2 \in U_2.$$

So, $(X \times U_1) \cap (X \times U_2) = X \times (U_1 \cap U_2) = \emptyset$ and

$(x_i, y_i) \in X \times U_i \quad (i=1, 2)$. And these are clearly open, by definition.

#12. Let X be a Hausdorff and $A \subseteq X$.

Given two distinct $x, y \in A$, there are opens $U, V \subseteq X$ s.t. $x \in U$, $y \in V$, $U \cap V = \emptyset$.

Now, $(U \cap A) \cap (V \cap A) = \emptyset$, containing x, y .

#13. X is $T_2 \Leftrightarrow \forall (x_1, x_2) \in X \times X, (x_1, x_2) \notin \Delta \Rightarrow \exists \underbrace{U, V}_{\text{open}} \subseteq X$ s.t. $(x_1, x_2) \in U \times V$
 $U \cap V = \emptyset \Leftrightarrow (U \times V) \cap \Delta = \emptyset$
 $\Leftrightarrow U \times V \subseteq \Delta^c$

$\Leftrightarrow \Delta$ is closed.

#14. $x = \lim x_n \Leftrightarrow$ Given open $U \subseteq \mathbb{R}$ containing x , there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow x_n \in U$.
 Since for any open $U \subseteq \mathbb{R}$, $\mathbb{R} \setminus U$ is finite, put $\# \mathbb{R} \setminus U = N$. Then,

x_n must be contained in U , for all $n \geq N$.

So, the answer is $\mathbb{R}$.

#19. Def: $\text{Bd } A = \overline{A} \cap \overline{X-A}$.

a). $\text{Int } A \cap \text{Bd } A = \emptyset$.

Proof. Suppose that $\text{Int } A \cap \text{Bd } A \neq \emptyset$. Let $x \in \text{Int } A \cap \text{Bd } A$.

Then, there is an open U containing x s.t. $x \in U \subseteq A$.

And, since $x \in \text{Bd } A = \overline{A} \cap \overline{X-A}$,

$$U \cap A \neq \emptyset \quad \text{and} \quad U \cap (X-A) \neq \emptyset$$

But $U \cap (X-A) \neq \emptyset \Leftrightarrow U \not\subseteq A$, so it is contradiction.

And, since $\text{Int } A, \text{Bd } A \subseteq \overline{A}$, $\text{Int } A \cup \text{Bd } A \subseteq \overline{A}$.

Conversely, for $x \in \overline{A}$, suppose that $x \notin \text{Int } A$. Then, there is an open U containing x s.t.

$$x \in U \quad \text{but} \quad U \not\subseteq A$$

$$U \not\subseteq A \Leftrightarrow U \cap (X-A) \neq \emptyset, \text{ so } x \in \text{Bd } A. \quad \square$$

b). $\text{Bd } A = \emptyset \Leftrightarrow \overline{A} = \text{Int } A$.

If A is clopen, $\overline{A} = A = \text{Int } A$. And, if $\overline{A} = \text{Int } A$, then

$$A \subseteq \overline{A} = \text{Int } A, \text{ so } A = \text{Int } A, \text{ and } \overline{A} = \text{Int } A \subseteq A, \text{ so } \overline{A} = A$$

Section 18.

#2. Let $V \subset Y$ be an open containing $f(x)$. Then,

$f^{-1}[V]$ is open of X , containing x .

Since x is the limit point of A , $f^{-1}[V] \cap A \neq \emptyset$. Now,

$$V \cap f[A] \supseteq f[f^{-1}[V] \cap A] = f[f^{-1}[V] \cap A] \neq \emptyset,$$

so $f(x)$ is the limit point of $f[A]$.

#6. $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$

f is continuous at only $x=0$.

#8.

Section 1a.

+ . $f: \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}; t \mapsto (t, t, \dots, t)$ is continuous under the product topology, but not continuous under the box topology.

Proof. Let $A = \prod_{n=1}^{\infty} (-\frac{1}{n}, \frac{1}{n})$ be given. Clearly, A is open of $\mathbb{R}^{\mathbb{N}}$ with box topology. But, $f^{-1}[A] = \{0\}$, which is not open of $\mathbb{R}$, so f is not continuous.

#6. Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence in $\prod_{\alpha \in \Lambda} X_\alpha$. Then,

A_n converges to A if and only if for each α , $\{\pi_\alpha(A_n)\}_{n \in \mathbb{N}}$ converges to $\pi_\alpha(A)$.

Proof. A_n converges to A if and only if

for any basis element $B \subseteq \prod_{\alpha \in \Lambda} X_\alpha$ containing A , there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow A_n \in B$

Suppose that $A_n \rightarrow A$. Given $\alpha \in \Lambda$, let $U_\alpha \subseteq X_\alpha$ be an open containing $\pi_\alpha(A)$.

Then, $B = \prod_{\alpha \in \Lambda} U_\alpha$ where $U_\beta = X_\beta$ if $\beta \neq \alpha$ is basis element containing A , so

there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow A_n \in B$. That is, $\pi_\alpha(A_n) \in U_\alpha = \pi_\alpha[B]$, $\forall n \geq N$.

Conversely, let $B \subseteq \prod_{\alpha \in \Lambda} X_\alpha$ be a basis element containing $A \in B$.

By definition, $B = \prod_{\alpha \in \Lambda} U_\alpha$ s.t. all $U_\alpha = X_\alpha$ but finitely many $\beta \in \Lambda$,

let such $\beta \in \Lambda$ as $\beta_1, \dots, \beta_k$. By the assumption, for each β_i , there is $N_i \in \mathbb{N}$ s.t.
 $n \geq N_i \Rightarrow \pi_{\beta_i}(A_n) \in U_{\beta_i}$.

Take $N = \max\{N_1, \dots, N_k\}$. Now,

$$n \geq N \Rightarrow A_n \in B. \quad \square$$

In box topology, $\Rightarrow$ holds but does not $\Leftarrow$.

$$\#7. \mathbb{R}^\infty = \bigcup_{n=1}^{\infty} \prod_{k=1}^n \mathbb{R} \times \underbrace{\mathbb{R} \times \dots \times \mathbb{R}}_n \times \underbrace{\{0\} \times \dots \times \{0\}}_n$$

$A \in \mathbb{R}^{\mathbb{N}}$ is contained in $\overline{\mathbb{R}^\infty}$ if and only if:

for any basis $\mathcal{U} \subseteq \mathbb{R}^{\mathbb{N}}$ containing A , $\mathcal{U} \cap \mathbb{R}^\infty \neq \emptyset$.

Under the Product Topology, for given $A \in \mathbb{R}^{\mathbb{N}}$, denote $\mathcal{U} \subseteq \mathbb{R}^{\mathbb{N}}$ as

$$\mathcal{U} = \bigcap_{n=1}^{\infty} \mathcal{U}_n \quad (\mathcal{U}_n = \{B \mid B \text{ all } n \text{ coord but finitely many } n_i \text{ and } \pi_n(A) \in \mathcal{U}_n, \forall n \in \mathbb{N}\})$$

$$\begin{aligned} \text{Then, } \mathcal{U} \cap \mathbb{R}^\infty &= \left(\bigcap_{n=1}^{\infty} \mathcal{U}_n \right) \cap \left(\bigcup_{n=1}^{\infty} \left(\left[\prod_{k=1}^n \mathbb{R} \right] \times \left[\prod_{k=n+1}^{\infty} \{0\} \right] \right) \right) \\ &= \bigcup_{n=1}^{\infty} \left[\left(\bigcap_{k=1}^{\infty} \mathcal{U}_k \right) \cap \left(\prod_{k=1}^{\infty} \chi_k^n \right) \right] \quad \text{where } \chi_k^n = \begin{cases} \{0\} & k > n \\ \mathbb{R} & k \leq n \end{cases} \\ &= \bigcup_{n=1}^{\infty} \bigcap_{k=1}^{\infty} (\mathcal{U}_k \cap \chi_k^n) \end{aligned}$$

It cannot be empty set because, if $m \in \mathbb{N}$ is a number s.t. $k \geq m \Rightarrow \mathcal{U}_k = \mathbb{R}$, then

$$\bigcap_{k=1}^{\infty} \mathcal{U}_k \cap \prod_{k=1}^{\infty} \chi_k^m \neq \emptyset$$

Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^{\mathbb{N}}$. But, under the box topology, it cannot be held, for

$$\mathcal{U} = \bigcap_{n=1}^{\infty} (0,1)$$

valid as counterexample. Actually, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$. (Becomes no such $A \in \overline{\mathbb{R}^\infty}$ with $A \notin \mathbb{R}^\infty$.)

#8. h is bijective, with the inverse as

$$h^{-1}: \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}: (x_n)_{n \in \mathbb{N}} \mapsto \left(\frac{1}{a_n} \cdot [x - b_n] \right)$$

And, regardless of box or prod. topol.

#10. Subbasis with Initial Topology.

Let X be a set, and $\{X_\alpha\}_{\alpha \in A}$ be a collection of spaces.

1. Given $f_\alpha: X \rightarrow X_\alpha$ ($\alpha \in A$), there exists a unique coarsest Topology on X s.t. for all $\alpha \in A$, f_α is continuous map.

Proof. Let $S_\alpha \stackrel{\text{def}}{=} \{f_\alpha^{-1}[U_\alpha] \mid U_\alpha \subseteq X_\alpha \text{ is open of } X_\alpha\}$

Then, $S \stackrel{\text{def}}{=} \bigcup_{\alpha \in A} S_\alpha$ is a subbasis of X , because: $X = f_\alpha^{-1}[X_\alpha]$, for any $\alpha \in A$.

Given $\alpha \in A$, and open $U_\alpha \subseteq X_\alpha$, $f_\alpha^{-1}[U_\alpha] \in S_\alpha \subseteq S$, so all f_α are continuous.

Moreover, it is the Coarsest Topology s.t. all f_α are continuous, because:

if $\mathcal{T}$ is a Topology on X satisfying such condition,

then for any $\alpha \in A$ and open $U_\alpha \subseteq X_\alpha$, $f_\alpha^{-1}[U_\alpha]$ is contained in $\mathcal{T}$, therefore $\mathcal{T}_S \subseteq \mathcal{T}$.

Note: Such a Topology on X is called the initial Topology for $\{f_\alpha\}$.

2. A map $g: Y \rightarrow X$ is continuous iff $f_\alpha \circ g$ is continuous for all $\alpha \in A$.

(The given Y is any space, and X is the initial Topology.)

Proof. Since the Composition of Continuous maps is Continuous $\Rightarrow$ is trivial.

Conversely, By the assumption, for each $\alpha \in A$, and an open $U_\alpha \subseteq X_\alpha$,

$$(f_\alpha \circ g)^{-1}[U_\alpha] = g^{-1}[f_\alpha^{-1}[U_\alpha]] \text{ is an open of } Y.$$

Meanwhile, by the first assertion, every open of X is generated by S ,

$$g^{-1}[U] \text{ is open for any open } U \subseteq X.$$

Section. Metric.

#2. Define $d: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R} : (x_1, y_1), (x_2, y_2) \mapsto \begin{cases} |y_1 - y_2| & \text{if } x_1 = x_2 \\ 1 + |y_1 - y_2| & \text{if } x_1 \neq x_2. \end{cases}$

Then, for any basis element of $\mathbb{R} \times \mathbb{R}$ with the dictionary order topology,

$((a, b), (c, d))$

$$B_r(x, y) = \begin{cases} \{(x, y) \mid |y_1 - y| < r\} & \text{if } r < 1 \\ \{(0, y)\} & \text{if } r \geq 1 \end{cases}$$

Compact space.

1. a). Let $\mathcal{C}, \mathcal{C}'$ be topologies on X , with $\mathcal{C} \subseteq \mathcal{C}'$.

If $(X, \mathcal{C}')$ is compact, then $(X, \mathcal{C})$ is compact.

Proof. Let $\mathcal{O} \in \mathcal{C}$ be an open cover of X . Then, since $\mathcal{O} \subseteq \mathcal{C}'$ and $(X, \mathcal{C}')$ compact, $\mathcal{O}$ has finite subcover. $\square$

1. b). If X is Compact T_2 under both $\mathcal{C}$ and $\mathcal{C}'$,

then either $\mathcal{C}$ and $\mathcal{C}'$ are equal or they are Not comparable.

Proof. If $\mathcal{C} = \mathcal{C}'$, then there is nothing to prove. Suppose $\mathcal{C} \neq \mathcal{C}'$.

Enough to show that neither $\mathcal{C} \subseteq \mathcal{C}'$ nor $\mathcal{C}' \subseteq \mathcal{C}$.

Suppose that $\mathcal{C} \subseteq \mathcal{C}'$. Then, we can choose an open $U \in \mathcal{C}' \setminus \mathcal{C}$.

Now, $X \setminus U$ is closed in $\mathcal{C}'$ but not $\mathcal{C}$. Since $\mathcal{C}'$ is Compact T_2 ,

$X \setminus U$ is Compact set of $\mathcal{C}'$. Since $\mathcal{C} \subseteq \mathcal{C}'$, for any open cover $\mathcal{O} \subseteq \mathcal{C}$ for $X \setminus U$,

it has finite subcover. Therefore, $X \setminus U$ is compact in $(X, \mathcal{C})$, thus closed.

But, it is contradiction. Similarly, $\mathcal{C}' \not\subseteq \mathcal{C}$.

2. a). Every subspace in Co-finite Topology $(\mathbb{R}, \mathcal{C}_f)$ is Compact.

b). In Co-Countable $(\mathbb{R}, \mathcal{C}_c)$, is $[0, 1]$ Compact? an

Proof. Let $A \subseteq \mathbb{R}$ be a subset. Let $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$ be open cover for A . There is,

$$A \subseteq \bigcup_{U \in \mathcal{O}} U \Leftrightarrow \bigcap_{U \in \mathcal{O}} \mathbb{R} \setminus U \subseteq A^c$$

But since $\mathbb{R} \setminus U$ is finite for each $U \in \mathcal{O}$, there are finite number of $U_i, \dots, U_n$ s.t.

$$\bigcap_{i=1}^n \mathbb{R} \setminus U_i \subseteq A^c \Leftrightarrow A \subseteq \bigcup_{i=1}^n U_i$$

Therefore A is Compact. (More specifically, for any distinct finite set F_1 and F_2 , $F_1 \cap F_2$ has number of less than $\min\{\#(F_1), \#(F_2)\}$.)

But, in the Co-Countable, $[0, 1]$ is Not compact, because

Construct sequences; denote $\mathcal{O} \cap [0, 1] = \{U_n \mid n \in \mathbb{N}\}$, and let $\mathcal{U}_n = (\mathbb{R} \setminus \mathbb{Q}) \cup \{x_n\}$

Then, $\bigcup_{n=1}^{\infty} U_n \supseteq [0, 1]$ but it has no finite subcover.

3. Finite Union of Compact Subspace is Compact.

Proof. Let $K_1, \dots, K_n \subseteq X$ be Compact subsets. Let $\mathcal{O} \subseteq \mathcal{P}(X)$ be an open cover of $\bigcup_{i=1}^n K_i$.

Since $\bigcup_{U \in \mathcal{O}} U \supseteq \bigcup_{i=1}^n K_i \supseteq K_j$ ($1 \leq j \leq n$), for each j , $\mathcal{O}$ has a finite subcover for K_j ,

denote $\mathcal{U}_j^1, \dots, \mathcal{U}_j^{r_j} \in \mathcal{O}$. Now, clearly

$$\bigcup_{i=1}^n K_i \subseteq \bigcup_{j=1}^n \bigcup_{m=1}^{r_j} \mathcal{U}_j^m$$

4. Every Compact Subspace in a Metric space is closed bounded.

Give the example that the Converse fails.

Proof. Closed is given by: Compact in T_2 is closed, because

Let X be a Hausdorff space, and $K \subseteq X$ be a Compact subset.

Let $x \in X \setminus K$ be fixed. Now, for each $y \in K$, there exists disjoint open $\mathcal{U}_y^x, \mathcal{U}_y$ s.t.

$$x \in \mathcal{U}_y^x, y \in \mathcal{U}_y, \mathcal{U}_y^x \cap \mathcal{U}_y = \emptyset$$

Now, $\{\mathcal{U}_y \mid y \in K\}$ is open cover of K , therefore there is a finite subcover

$\{\mathcal{U}_{y_i} \mid i=1, 2, \dots, r_x\}$ and open neighborhoods of x $\{\mathcal{U}_{y_i}^x \mid i=1, \dots, r_x\}$.

Put $V_x = \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}^x$. Then, V_x is open neighborhood and $K \cap V_x = \emptyset$, thus $V_x \subseteq K^c$.

(More specifically, $K \subseteq \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}$ and $\left(\bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}\right) \cap V_x = \bigcup_{i=1}^{r_x} (\mathcal{U}_{y_i} \cap V_x) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset$).

And boundedness is clear.

Counterexample: Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$, discrete topology.

Then, it is closed itself, and bounded, but not Compact.

15. Let X be a Hausdorff, and $A, B \subseteq X$ be disjoint compact subsets.

Then, there exist disjoint open sets U and V containing A and B , respectively.

Proof. Let $x \in A$ be fixed. Then, for each $y \in B$, there exist opens $U_{(x,y)}$ and $V_{(x,y)}$ s.t.

$$x \in U_{(x,y)}, \quad y \in V_{(x,y)}, \quad U_{(x,y)} \cap V_{(x,y)} = \emptyset.$$

Since $\{V_{(x,y)} \mid y \in B\}$ is open cover of B , there exists a finite subcover

$$\{V_{(x,y_i)} \mid i=1, 2, \dots, r_x\}. \text{ Now, } U_x = \bigcap_{i=1}^{r_x} U_{(x,y_i)} \text{ is open neighborhood of } x.$$

$$V_x = \bigcup_{i=1}^{r_x} V_{(x,y_i)} \text{ is open set containing } B, \text{ and } U_x \cap V_x = \emptyset. \text{ because}$$

$$U_x \cap V_x = U_x \cap \left(\bigcup_{i=1}^{r_x} V_{(x,y_i)} \right) = \bigcup_{i=1}^{r_x} (U_x \cap V_{(x,y_i)}) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset$$

In summary, for each $x \in A$, there are two disjoint opens U_x, V_x s.t.

$$x \in U_x, \quad B \subseteq V_x, \quad U_x \cap V_x = \emptyset.$$

And, $\{U_x \mid x \in A\}$ is an open subcover of A , thus it has a finite subcover

$$\{U_{x_i} \mid i=1, \dots, m\}. \text{ Finally,}$$

$$\bigcup_{i=1}^m U_{x_i} \text{ and } \bigcap_{i=1}^m V_{x_i} \text{ are disjoint open sets containing } A \text{ and } B, \text{ respectively.}$$

$$\text{Disjointness is given by } \left(\bigcup_{i=1}^m U_{x_i} \right) \cap \left(\bigcap_{i=1}^m V_{x_i} \right) = \bigcup_{i=1}^m \left(U_{x_i} \cap \left(\bigcap_{i=1}^m V_{x_i} \right) \right) = \bigcup \emptyset = \emptyset.$$

16. If $f: X \rightarrow Y$ is continuous map where X is compact and Y is Hausdorff, then f is closed map.

Proof. Let $C \subseteq X$ be a closed set. Since X is compact, C is compact.

And, the continuous map f preserves compactness, $f[C]$ is compact in T_2 , thus closed.

17. If Y is compact, then the projection $p_X: X \times Y \rightarrow X$ is a closed map.

Proof. Let $C \subseteq X \times Y$ be a closed set. Want to show that $X \setminus p_X[C]$ is open.

Let $x \in X \setminus p_X[C]$ be given. Then, $\{x\} \times Y \cap C = \emptyset$, therefore $\{x\} \times Y \subseteq (X \times Y) \setminus C$.

Now, by the Tube Lemma there is an open $U \subseteq X$ s.t.

$$\{x\} \times Y \subseteq U \times Y \subseteq (X \times Y) \setminus C.$$

Now, $x \in U$ and $U \subseteq X \setminus p_X[C]$, thus $X \setminus p_X[C]$ is open.

Note: the converse is true, but so hard to prove.

Recall: The Tube Lemma.

Let X be a topological space, and Y be a compact space.

Then, for the product space $X \times Y$, for fixed $x_0 \in X$,

For any open $N \subseteq X \times Y$ s.t. $\{x_0\} \times Y \subseteq N$, there is open $W \in \mathcal{C}_X$ s.t. $\{x_0\} \times Y \subseteq W \times Y \subseteq N$.

Proof. Since $\{x_0\} \times Y \rightarrow Y: (x_0, y) \mapsto y$ is homeo, $\{x_0\} \times Y$ is compact.

Since N is open, for any $y \in Y$, $\exists U_y \in \mathcal{C}_X, V_y \in \mathcal{C}_Y$ s.t. $\{x_0\} \times Y \subseteq U_y \times V_y \subseteq N$.

Now, open cover $\{U_y \times V_y \mid y \in Y\}$ of $\{x_0\} \times Y$ has a finite subcover

$$\{U_{y_i} \times V_{y_i} \mid i=1, \dots, N\}.$$

Set $W = \bigcap_{i=1}^N U_{y_i}$. Then, $\{x_0\} \times Y \subseteq W \times Y$ is clear, and

$$(x, y) \in W \times Y \Leftrightarrow \forall i=1, \dots, N, x \in U_{y_i} \text{ and } \exists y \in V_{y_i}$$

$$\Rightarrow (x, y) \in U_{y_i} \times V_{y_i} \subseteq N.$$

8. Let X be any space, Y be a Compact Hausdorff. Then,

$f: X \rightarrow Y$ is continuous $\xleftrightarrow{T_2}$ if and only if $G_f = \{(x, f(x)) \mid x \in X\} \subseteq X \times Y$ is closed.

Proof. Suppose that f is continuous. Let $(x, y) \in (X \times Y) \setminus G_f$ be given. Clearly $y \neq f(x)$. Since Y is T_2 , there exist disjoint opens $U, V \subseteq Y$ containing $y, f(x)$, respectively.

In other words, $(y, f(x)) \in U \times V \subseteq (Y \times Y) \setminus \Delta(Y)$

And $f(x) \in V$ implies $x \in f^{-1}[f[x]] \subseteq f^{-1}[V] \subseteq X$. Now enough to show that only (x, y)
 $(x, y) \in f^{-1}[V] \times U \subseteq (X \times Y) \setminus G_f$

because f is continuous, thus $f^{-1}[V]$ is open. The inclusion (*) equals to:

$$(f^{-1}[V] \times U) \cap G_f = \emptyset$$

and it is true, if it is not empty set, then for some $(x, f(x)) \in G_f$,
 $(x, f(x)) \in f^{-1}[V] \times U$. That is, $f(x) \in V$ and $f(x) \in U$, which is contradiction.

(Note: in this direction we used only T_2 condition.)

Conversely, suppose that the graph G_f is closed set.

Let $U \subseteq Y$ be an open set. Want to show that: $f^{-1}[U]$ is an open set of X .

Let $x \in f^{-1}[U]$ be given. Then, $f(x) \in U$ by definition.

Let $x_0 \in X$ be given. Then, for any open $V \subseteq Y$ containing $f(x_0)$,

$$G_f \cap (X \times (Y \setminus V))$$

is closed set. Now, since Y is compact, the projection $\pi_X: X \times Y \rightarrow X$ is closed map, thus

$$\pi_X[G_f \cap (X \times (Y \setminus V))] \subseteq X \text{ is closed in } X.$$

Now, $U = X \setminus \pi_X[G_f \cap (X \times (Y \setminus V))]$ is open, moreover, $x_0 \in U$ because

$$x_0 \in U \Leftrightarrow x_0 \notin \pi_X[G_f \cap (X \times (Y \setminus V))] \Leftrightarrow \nexists (x, y) \in G_f \cap (X \times (Y \setminus V)), \pi_X(x, y) = x \neq x_0$$

But, since $(x_0, f(x_0)) \notin (X \times (Y \setminus V))$, the last assertion holds. Now, only showing $f^{-1}[U] \subseteq V$ is remained.

Let $x \in U$. Then, $f(x) \notin Y \setminus V$, that is, $f(x) \in V$. ~~is~~

79. The Generalized Tube Lemma.

Let $A \subseteq X$, $B \subseteq Y$ be compact subspaces. Then,

For any open $N \subseteq X \times Y$ containing $A \times B$, there exists opens $U \subseteq X$ and $V \subseteq Y$ s.t. $A \times B \subseteq U \times V \subseteq N$

Proof.

72. Let $p: X \rightarrow Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}[\{y\}]$ is compact.

If Y is compact, then X is compact. (Such p is called perfect map.)

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{y \in Y} \{y\}\right] = \bigcup_{y \in Y} p^{-1}[\{y\}]$$

Let $\mathcal{O} = \{U_\alpha \mid \alpha \in \Lambda\}$ be an open cover of X . For each $y \in Y$,

$$p^{-1}[\{y\}] \subseteq \bigcup_{y \in Y} p^{-1}[\{y\}] \subseteq \bigcup_{\alpha \in \Lambda} U_\alpha$$

And by the assumption, $p^{-1}[\{y\}]$ is compact, therefore there is a finite index $\Lambda_y \subseteq \Lambda$ s.t.

$$p^{-1}[\{y\}] \subseteq \bigcup_{i \in \Lambda_y} U_i.$$

Now, since p is closed map,

$$V_y = Y \setminus p[X \setminus (\bigcup_{i \in \Lambda_y} U_i)]$$

is an open set of Y , containing y . Now, since Y is compact, for some $y_1, \dots, y_n$,

$$Y = \bigcup_{y \in Y} V_y = \bigcup_{j=1}^n V_{y_j} \Rightarrow X = p^{-1}[Y] = p^{-1}\left[\bigcup_{j=1}^n V_{y_j}\right] = \bigcup_{j=1}^n p^{-1}[V_{y_j}]$$

Meanwhile, $p^{-1}[V_{y_j}] = p^{-1}[Y \setminus p[X \setminus (\bigcup_{i \in \Lambda_{y_j}} U_i)]] = X \setminus p^{-1}[p[X \setminus (\bigcup_{i \in \Lambda_{y_j}} U_i)]] \subseteq \bigcup_{i \in \Lambda_{y_j}} U_i$,

therefore $X \subseteq \bigcup_{j=1}^n \bigcup_{i \in \Lambda_{y_j}} U_i$

Γ + Extra problem

#1. Let X be a Hausdorff space, and $K_\alpha \subseteq X$ ($\alpha \in \Lambda$) be compact subsets. If $\{K_\alpha \mid \alpha \in \Lambda\}$ satisfies F.I.P., then $\bigcap_{\alpha \in \Lambda} K_\alpha \neq \emptyset$.

Proof. Suppose that $\bigcap_{\alpha \in \Lambda} K_\alpha = \emptyset$. Let $\beta \in \Lambda$ be fixed. Then,

$$K_\beta \cap \left(\bigcap_{\alpha \neq \beta} K_\alpha \right) = \bigcap_{\alpha \in \Lambda} K_\alpha = \emptyset, \text{ thus } K_\beta \subseteq \bigcup_{\alpha \neq \beta} K_\alpha^c$$

Since X is T_2 , K_α^c is open, so $\{K_\alpha^c \mid \alpha \in \Lambda \setminus \{\beta\}\}$ is open cover of K_β . Now, there is a finite subcover so

$$K_\beta \subseteq \bigcup_{i=1}^n K_{\alpha_i}^c, \text{ that is, } K_\beta \cap \left(\bigcap_{i=1}^n K_{\alpha_i} \right) = \emptyset.$$

It is contradiction to F.I.P.

#2. Let X be a compact set, and $A \subseteq X$ be an infinite subset. Then, $A' \neq \emptyset$.

Proof. Suppose that $A' = \emptyset$. Then, for any $x \in X$, there is an open U_x containing x

$$A \cap U_x \setminus \{x\} = \emptyset.$$

Since $\{U_x \mid x \in X\}$ is open cover of X , there is a finite subcover

$$\{U_{x_i} \mid i=1, 2, \dots, n\}.$$

Then, $A \subseteq X = \bigcup_{i=1}^n U_{x_i}$, but $A = A \cap \left(\bigcup_{i=1}^n U_{x_i} \right) = \bigcup_{i=1}^n (A \cap U_{x_i}) \subseteq \bigcup_{i=1}^n \{x_i\}$, which is contradiction.

#3. Co-Countable space $(\mathbb{R}, \mathcal{C}_c)$.

In $(\mathbb{R}, \mathcal{C}_c)$, $A \subseteq \mathbb{R}$ is Compact $\Leftrightarrow A$ is finite.

Proof. If A is finite, then trivial. If A is infinite, construct $\{x_n\}$ as

$x_1 \in A$, $x_2 \in A \setminus \{x_1\}$, $x_3 \in A \setminus \{x_1, x_2\}$, ..., then,

$U_n = \mathbb{R} \setminus \{x_i \mid i \in \mathbb{N} \setminus \{n\}\}$ is open of the given space, and

$A \subseteq \bigcup_{n \in \mathbb{N}} U_n$ but, it has no finite subcover

If $f: \mathbb{R} \rightarrow (\mathbb{R}, \mathcal{C}_c)$ is continuous, it is Cont. map.

Proof. Since the Compactness is preserved by Continuous, for any $[a, b]$,

$f[[a, b]]$ is compact in $(\mathbb{R}, \mathcal{C}_c)$.

That is, $f[[a, b]]$ is finite for any $a < b$. Now,

$f[\mathbb{R}] = f\left[\bigcup_{n \in \mathbb{Z}} [n, n+1]\right] = \bigcup_{n \in \mathbb{Z}} f[[n, n+1]]$ is at most Countable.

Meanwhile, since the Closedness is preserved by Continuous Preimage,

every finite or countable subset C , $f^{-1}[C]$ is closed.

Denote $\text{Im } f = \{x_n \mid n \in \mathbb{N}\}$. Let $A = f^{-1}[\{x_1\}]$, $B = f^{-1}[\{x_n \mid n \in \mathbb{N} \setminus \{1\}\}]$.

Then, $\mathbb{R} = A \cup B$ and $A \cap B = \emptyset$. And both A and B are closed,

$A = \mathbb{R} \setminus B$ is clopen, $A = \mathbb{R}$. Contradiction.

* Fail Try of 12.

T2. Let $p: X \rightarrow Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}[\{y\}]$ is compact.

If Y is compact, then X is compact. (Such p is called perfect map.)

Proof. Let $\mathcal{C} \subseteq \mathcal{C}_X$ be an open cover of X . That is, $\bigcup_{U \in \mathcal{C}} U = X$
Now, by the surjectivity, $p\left[\bigcup_{U \in \mathcal{C}} U\right] = \bigcup_{U \in \mathcal{C}} p[U] = Y$.

And, since p is closed onto map, it is open map. Therefore, $p[U]$ ($U \in \mathcal{C}$) is open cover of Y , thus it has finite subcover s.t. $\bigcup_{i=1}^N p[U_i] = Y$, since Y is compact. Observe that

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{i=1}^N p[U_i]\right] = \bigcup_{i=1}^N p^{-1}[p[U_i]]$$

Let $y \in Y$ be given. By the surjectivity, $p^{-1}[\{y\}]$ is non-empty.

Let $x_0 \in X$ s.t. $p(x_0) = y$.

$$\begin{aligned} p^{-1}[W] &\subseteq U \\ \Leftrightarrow W &\subseteq p[U] \end{aligned}$$

Let $y \in Y$ be given. If $U \subseteq X$ is open containing $p^{-1}[\{y\}]$, then

$$p^{-1}[\{y\}] \subseteq U \Rightarrow p[p^{-1}[\{y\}]] \subseteq p[U] \Rightarrow y \in p[U]$$

$$\Rightarrow y \in W \subseteq p[U]$$

$$Y = \bigcup_{y \in Y} (Y \setminus p[X \setminus (\bigcup_{i \in \mathbb{N}} U_i)]) = \bigcup_{j=1}^n (Y \setminus p[X \setminus (\bigcup_{i \in \mathbb{N}_j} U_i)])$$